Fiscal Year:	FY 2022	**Task Last Updated:**	FY 06/24/2022
PI Name:	Wyatt, Sarah E Ph.D.		
Project Title:	Spaceflight Alters Post-Transcriptional Regulation		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Zip Code:	45701	**Congressional District:**	15
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG. App B: Flight and Ground Space Biology Research
Start Date:	09/01/2019	**End Date:**	05/31/2023
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA KSC
Contact Monitor:	Zhang, Ye	**Contact Phone:**	321-861-3253
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 05/31/2023 per NSSC information (Ed., 6/6/22).		
Key Personnel Changes/Previous PI:	The postdoctoral fellow accepted a new position at Kennedy Space Center.		
COI Name (Institution):	Perera, Imara Ph.D. (North Carolina State University)		
Grant/Contract No.:	80NSSC19K1481		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Plants will be a crucial component for astronaut health and well-being during any long-distance spaceflight or colonization mission: as a source of food, for replenishing water and purifying air as well as physiological and psychological comfort. The challenge is to understand how plants respond to the spaceflight environment to enable plants to thrive in potentially hostile environments. As technologies have advanced for global gene expression, transcriptome level research has become an integral part of experiments and provides knowledge of gene expression resulting from the conditions of spaceflight. Steady state transcript abundance quantified by RNASeq is frequently used as measure of gene expression with the implicit assumption that transcriptional changes upon a treatment (such as a stress) are indicative of the downstream response. However, changes in mRNA abundance do not necessarily lead to corresponding changes in protein, and transcript and protein abundance are not highly correlated. The data from spaceflight experiments Biological Research in Canisters-20 (BRIC20) (Principal Investigator: Wyatt) and Plant Signaling (Principal Investigator: Perera) suggest that post-transcriptional regulation plays an integral role in gene expression differences between spaceflight and ground controls. The differential expression, abundance, and phosphorylation of translational machinery in spaceflight leads to the obvious questions: What genes are being post transcriptionally regulated and by what mechanism(s)? To answer these questions, principal investigators Wyatt and Perera plan to combine expertise and resources to generate compatible multi-omics datasets and provide a comprehensive picture of transcriptional and post transcriptional regulation. This integrated approach will help answer fundamental questions regarding plant adaptation to spaceflight and this proposal is aligned with sub-topic PL-A of Appendix B ("Study plant growth in spaceflight environments and physiological responses to the multiple stimuli encountered in those environments"). The novel datasets and analyses will address questions outlined in Research Emphasis 2 of the Space Biology Science Plan 2016-2025 (Plant Biology Studies in Support of Space Exploration), specifically "(a) Answer basic questions about how plants respond to changes in gravity and other environmental factors associated with spaceflight and (b) Build on past plant research to provide a better understanding of physiological responses of plants to spaceflight, providing new knowledge that can facilitate the development of a bioregenerative life support system." Analyses will also address Research Emphasis 4: molecular and cellular biology, specifically "(a) Studies designed to determine how spaceflight alters gene expression at the transcriptomic, metabolomics and proteomics levels in the different tissues or cell types within the same organism, and how these changes impact the organism's overall health during space travel."
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Plants are critical for human life and well-being on Earth. However, many areas of the world have limited-resources for adequate plant growth to support life. For plants to thrive in these less than hospitable environments, the challenge is to understand how plants respond to a variety of stresses and be able to mitigate those responses. Researchers often use the transcriptome, those genes transcribed from DNA, as a proxy for protein produced because of the speed and relative ease of acquisition and analysis. Full analysis of proteins requires much more tissue and time but also only provides a subsection of the protein complement due to current limitation of techniques. Here, we use analysis of polysome associated RNAs, those transcripts that are associated with polysomes, the translational machinery of the cell. The research proposed will also generate datasets for the full transcriptome and small, regulatory RNAs. The goal is to use the three datasets to provide a more comprehensive picture of gene expression and its regulation as affected by spaceflight. The project will answer basic questions about how plants respond to changes in gravity and other environmental factors associated with spaceflight, and build on past plant research to provide a better understanding of physiological responses of plants to spaceflight, providing new knowledge that can facilitate the development of a bioregenerative life support system. In addition, it will also add to our understanding of how spaceflight alters gene expression at the gene transcript and protein levels in both roots and shoots. Overall, it will lead to findings that may provide potential targets for environmental or genetics adaptions for plant growth for food and other uses during spaceflight impacting astronaut help and well-being. In addition, it will provide a deeper understand of how plants respond to environmental factors that may also benefit plant growth on Earth on marginal lands or in limited spaces to increase production.
	Summary of work in Year 3 – Fight, data acquisition and analyses: Year 1 of the project saw the completion of several of the preflight objectives on multiple avenues of inquiry. Year 2 focused on preflight verification experiments Science (S) and Experiment (E) Verification Tests (VT), and launch readiness reviews and final launch preparation, culminating in launch aboard SpaceX-22 (SpX-22) on June 3, 2021. Year 3 saw the partial return of the flight samples, leading to discussion with Kennedy Space Center about options and ultimately a second opportunity to fly the experiment. APEX-07 original flight and return – June-July 2021: APEX-07 launched aboard SpaceX-22 on June 3, 2021, and plates were inserted into the Veggie units on June 14, 2021. On June 26, 2021, 12 days after insertion, the samples were removed from the Veggie units, and placed at -80C. On July 9, 2021, samples were transferred from the on-orbit freezer (Glacier) to a return freezer (Polar) in the Dragon capsule and returned to Earth. Unfortunately, only 35 plates made it into the Polar freezer and made the trip home; the other 25 were left in the Glacier, which had been powered down and were discovered 14h later, having been at ambient temperature aboard the International Space Station (ISS) for the duration. These were a total loss and not returned. On June 28, 2021, the ground controls were harvested, using the same protocol, at NASA Kennedy Space Center (KSC) by Susan Manning-Roach, assisted by Gerard Newsham. Flight samples and their corresponding ground controls were shipped on July 12 to the respective institutions for analyses: 16 (randomly pulled from Veggie 1) to North Carolina State University (NCSU), 6 (3 from Veggie A and 3 from B) to Texas A&M University (TAMU), with the remaining 13 (3 from B and 10 from A) to Ohio University (OU). The "extra" control samples were shipped to OU. Making lemonade out of lemons (hopefully): Dr. Dorothy Shippen (TAMU), a world expert in plant telomere regulation, had contacted the GeneLab Plant Analysis Working Group (of which Dr. Perera and I are members) to see if any researcher had "extra" spaceflight samples for analysis of telomere length and regulation in plants. Obviously, not a common occurrence. But timing is everything. With the accidental loss of the spaceflight samples, and our inability to use the samples returned to accomplish our experiments, we provided Dr. Shippen with 3 plates each from flight and ground controls. This has been a productive collaboration with exciting results and no cost incurred to us or NASA. A publication is in the last stages of preparation prior to submission in July. Dr. Perera and I are working to find resources to use the rest of the flight samples to enrich our analysis of post transcriptional regulation.

Task Progress:	Preparation for second flight – December 14-21, 2022: Plating, growth, and collection. Flight preparation was performed at the Space Station Processing Facility at Kennedy Space Center during the week of December 14, 2021. All methods were performed exactly as they were for the previous flight on SpaceX-22. Prior to arrival at Kennedy Space Center, polyethersulfone (PES) membranes were cut to fit 10 cm square Petri dishes and sterilized by autoclave. Arabidopsis 35S::HF-RPL18 seeds were sterilized using 30% bleach, allowed to air dry, and adhered to the PES membranes with 1% guar gum, 15 seeds per membrane, 60 membranes total. Once at KSC, 0.8% agar was prepared with ½ MS solution and loaded into sterile 10 cm square Petri dishes, 40 mL each. The agar plates were stored at 4°C overnight. For final set-up of the plates for the experiment, seeded membranes were laid on the agar in Petri dishes (December 15, 2021), plates sealed with micropore tape, and wrapped in aluminum foil, 10 plates per sleeve, and stored at 4°C until turnover to Cold Stowage on December 19, 2021. The refly of APEX-07 launched on SpaceX-24, December 21, 2021. The payload arrived at the ISS on December 23 and remained at 4°C until activation. On December 28, 2021, to initiate the experiment, 30 plates were inserted into each of the two Veggie units aboard the ISS. Lighting was set at long day conditions (16 h light; 8 h dark) using 100µmol/m2/s of red (630 nm), blue (455 nm) and green (530 nm) light, as is standard for Veggie. On January 10, 2022, 12 days after insertion, the samples were removed from the Veggie units, and placed at -80C. On January 23, 2021, samples were transferred from the on-orbit freezer (Glacier) to a return freezer (Polar) in the Dragon capsule and returned to Earth. The same protocol was followed for the ground controls, starting with preparation of agar on December 20 and seeding of plates on December 21, 2021. Plates were stored at 4°C until insertion into two Veggies units at Kennedy Space Center on January 3, 2022. Temperature and lighting conditions were set to replicate conditions with those on the ISS. Controls were harvested January 16, 2022 and frozen at -80°C until return to investigators. Ohio University – mRNA and small RNA processing: Samples were shipped to Ohio University on dry ice in January 2022. Once received, the samples were carefully unpacked and stored at -80°C until processing. Plates were processed individually for RNA isolation. Each plate was removed from -80°C and unwrapped from its protective foil. Shoot material was dissected from root material, and each tissue was separately pulverized using a mortar and pestle with liquid nitrogen. RNA extractions were performed using 30-50 µg of each tissue and the remainder of the material was frozen for future use. Total RNA was extracted using the Norgen Plant/Fungi Total RNA Purification Kit. RNA was isolated separately from root and shoot tissue, isolated individually from four plates each from flight and ground controls, resulting in 32 total RNA samples. RNA integrity was assessed via the Agilent 2100 Bioanalyzer. RNA yields were 15-30 ng per extraction and all RNA integrity number (RIN) values were above 8; thus, yield and quality metrics met or surpassed targets for APEX-07 RNA isolation. North Carolina State University – TRAPseq Processing of APEX-07 Reflight Samples: The reflight samples were received at NCSU at the end of January 2022. Once received, the samples were carefully packed and stored at -80°C until processing. During the months of April and May, the new set of reflight space and ground samples were processed for polysome associated RNAs. For optimal workflow, the Co-Principal Investigator (CoPI) and graduate student worked together to separate the frozen tissue into root and shoot samples, and needed to pool several plates to obtain sufficient mass for the downstream processing. We also established a pipeline of processing 4-6 samples/day. Overall, polysome associated RNA from shoot samples was excellent, with average yields of ~ 6 µg/TRAP preparation. Root yields were ~ 1.5-2 µg/preparation. The library preparation for RNA-Seq requires ~ 1 µg, and we have easily met this requirement. NCSU Library Preparation and Sequencing (all samples): Total RNA samples (received from Ohio University) and polysomeassociated RNA samples have been submitted to the Genomics Sciences Laboratory at NCSU for library preparation and sequencing. We have four replicate samples each of root and shoot tissue from spaceflight and ground controls for both total and polysomal RNA for RNA-Seq (total of 32 samples), as well as 16 total RNA samples for sRNA library preparation and sequencing. Data should be returned in ~ 8 weeks.
Bibliography Type:	Description: (Last Updated: 11/25/2025)
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